GATE-2001

BT: BIOTECHNOLOGY

Directions (Q. 1 – 25): *For each sub-question given below, four alternatives are provided of which only one is correct. Write the correct answer in the answer book by writing (a), (b), (c) or (d) along with the corresponding sub-question number. (25 × 1 = 25)*

1. The limiting factor in the production of large quantities of ethanol as bio-fuel is
   (a) Lack of a balanced medium
   (b) Ethanol toxicity to cells
   (c) Expensive downstream processing steps
   (d) Only (b) and (c) of the above

1. Gel retardation assay is routinely used to monitor interactions between
   (a) Proteins
   (b) Drug and nucleic acid
   (c) Nucleic acids
   (d) Protein and nucleic acid

3. Efficient expression of a heterologous protein product is influenced by
   (a) Transcriptional efficiency
   (b) Copy number of the plasmid
   (c) Codon bias
   (d) All of these

4. The fundamental feature of the genetic code which allows the expression of a protein in any host is its
   (a) Triplet nature (b) Universality
   (c) Degeneracy (d) Redundancy

5. When thymidine synthesis is inhibited, eukaryotic cells will be arrested at
   (a) End of S phase (b) End of M phase
   (c) G1/S interphase (d) G2/M interphase

6. To PCR amply the sequence
   ATCTTCTACG.....AAGCTTCCGG
   TAGAAGATGC.....TTCGAACGCC
   the required primers are
   (a) ATCTTCTA and CGAACGCC
   (b) ATCTTCTA and CCGCAAGC
   (c) TAGAAGAT and CGAACGCC
   (d) TAGAAGAT and CCGCAAGC

7. The major groove of DNA is lined by
   (a) N2 of purine and N1 of pyrimidine
   (b) N7 of purine and O2 of pyrimidine
   (c) N7 of purine and C4 of Pyrimidine
   (d) None of these

8. The heterozygosity of any locus can be ascertained by
   (a) RFLP analysis
   (b) SNPs
   (c) FISH analysis
   (d) Either RFLP analysis or SNPs

9. BAC, which can be used to clone large DNA fragments, is derived from
   (a) Col E plasmid (b) F plasmid
   (c) 2μ (d) Mu phage

10. Antibody diversity is generated by
    (a) Protein splicing
    (b) Somatic mutation
    (c) Allelic exclusion
    (d) Inter-chromosomal recombination

11. Resistance to herbicide chlorsulfuron in plants is due to a change in
    (a) Glutamine synthetase
    (b) Threonine deaminase
    (c) Acetolactate synthase
    (d) DNA polymerase

12. A heterologous protein for its expression in the milk of a transgenic animal should be under the control of the promoter of the gene coding for
    (a) β-Globin (b) β-Lactoglobulin
    (c) Preproinsulin (d) *lac* Z

13. To isolate a gene coding for glucagon, the cDNA library has to be constructed using mRNA isolated from
    (a) Intestine (b) Pancreas
    (c) Pituitary (d) Brain

14. If the fractional recovery at each step of unit operation is 0.8, the recovery after 4 steps will be
    (a) 0.24 (b) 3.23
    (c) 0.41 (d) 0.82

15. The factor (s) likely to increase the rate of reaction catalyzed by a surface immobilized enzyme is/are
(a) Increased agitation of the bulk liquid containing the substrate
(b) Continued replacement of the bulk liquid containing the substrate
(c) Increased concentration of the substrate in the bulk liquid
(d) All of these

16. Which of the following cases are likely to lead to faster rates of catalysis by an enzyme immobilized on a negatively charged support ?
(a) A positively charged substrate and a negatively charged product
(b) A negatively charged substrate and a positively charged product
(c) A positively charged substrate and a positively charged product
(d) None of the above

17. The essential component of Ti plasmid required for integration into plant genome is
(a) Origin of replication
(b) Tumor inducing gene
(c) Nopaline utilization gene
(d) All of these

18. Positional cloning approach exploits information
(a) On the location of the gene in the genome
(b) On the status of its expression
(c) About the position of promoter relative to MCS
(d) About the position of the restriction sites

19. Hormone pairs required for a callus to differentiate are
(a) Auxin and cytokinin
(b) Auxin and gibberellin
(c) Ethylene and gibberellin
(d) Cytokinin and gibberellin

20. Reverse genetics means
(a) Finding the function of a ORF
(b) Finding the gene responsible for a trait
(c) RNA dependent DNA synthesis
(d) Converting somatic cell to a germ cell

21. Which of the following statements applies to the operation of a fed-batch process ?
(a) The volume of the culture increases with time
(b) It helps controlling repressive effects of the nutrient being fed
(c) It eliminates the need for oxygen supply
(d) Only (a) and (b) of the above

22. A gene can not be isolated from a human genomic DNA library by functional complementation in *E. coli* because of
(a) Non-functional promoter
(b) The absence of splicing machinery
(c) Coupled transcription and translation
(d) Codon bias

23. In large-scale fermentation, the preferred method of sterilization is
(a) Chemicals (b) Radiation
(c) Filtration (d) Heat

24. Embryo rescue is a useful technique to
(a) Grow/generate hybrids between different plant species
(b) Complete the growth of embryos susceptible to defects in seed development
(c) Break the dormancy of seeds
(d) All of the above

25. Which of the following is not true of aerobic digestion ?
(a) It generates most sludge
(b) It generally incurs higher running cost
(c) It may generate a usable fuel
(d) Requires a shorter residence time

26. Match the products in **Column A** with their corresponding organisms in **Column B**

Column A	**Column B**
(a) *Aspergillus niger*	1. Lysine
(b) *Saccharomyces cerevisiae*	2. Citric acid
(c) *Penicillium chrysogenum*	3. Acetone/butanol
(d) *Lactobacillus casei*	4. Vitamin B_{12}
(e) *Corynebacterium glutamicum*	5. Erythromycin
	6. β-Lactam
	7. Diacetyl
	8. Ethanol

27. Match the unit operations in **Column A** with the most appropriate recovery stage in **Column B**

Column A	**Column B**
(a) Drying	1. Pretreatment
(b) Sedimentation	2. Purification
(c) Membranes	3. Formulation
(d) Cell disruption	4. Solid/liquid separation
(e) Chromatography	5. Solid/solid separation
	6. Concentration

28. (*a*) A chemostat is operating at steady state at a dilution rate of 0.1 hr^{-1} and a limiting nutrient concentration of 10 μM. If μ_{max} for the organism is 0.5 hr^{-1}. Calculate the Monod constant for the nutrient. (3)

(*b*) Why asparaginase is used in anticancer therapy? (2)

29. (*a*) Write the reaction catalyzed by penicillin G acylase. (1)

(*b*) Name any two techniques by which penicillin G acylase may be immobilized. (2)

(*c*) Why are mammalian cells cultured in CO_2 incubators? (1)

(*d*) Mention an important post-translational modification absent in prokaryotes making them unsuitable hosts for expressing human genes. (1)

30. (*a*) The sequences at the cloning site of three vectors are given below. The *Bam* HI (GGATCC) and *Hind* III. (AAGCTT) sites are underlined. Only the sequences around the restriction sites are shown. The symbol "_____" indicates rest of the sequence. (1)

Vector 1: _____ Promoter ... ATGGGTCGCGGATCCGGCTGC .. AAGCTT _____

Vector 2 : _____ Promoter ... ATGGGTCGGGATCCGGCTGCT .. AAGCTT _____

Vector 3 : _____ Promoter ... ATGGGTCGGATCCGGCTGCTA.. AAGCTT _____

Which one of the above three vectors is appropriate to clone the following ORF

_____ ATGCCCAACACCCGGATCCCG.. TAAAAGCTT _____

for expression? Give the reason in one sentence.

(*b*) Draw the restriction map of the plasmid given the following data (the gel pattern shown below is not to scale). The size of each DNA fragment (in kb) is indicated next to it (3)

EcoRI	SalI	HindIII	EcoRI & HindIII	SalI & HindIII	EcoRI & SalI
5.4 –	5.4 –				
		2.1 –	2.1 –		3.6 –
		1.9		1.9 –	
		1.4 –	1.4 –	1.4 –	1.8 –
			1.3 –	1.2 –	
			0.6 –	0.9 –	

ANSWERS

1. (*d*)	2. (*d*)	3. (*d*)	4. (*c*)	5. (*a*)	6. (*b*)	7. (*c*)	8. (*d*)	9. (*b*)	10. (*c*)
11. (*c*)	12. (*b*)	13. (*b*)	14. (*c*)	15. (*b*)	16. (*a*)	17. (*b*)	18. (*a*)	19. (*a*)	20. (*a*)
21. (*d*)	22. (*b*)	23. (*d*)	24. (*d*)	25. (*c*)					

Note : Q. No. 26 to 30 are Subjective Questions.

EXPLANATIONS

1. For ethanol commercial utilization we need high purity which leads to heavy downstream processing cost along with high percentage of ethanol cause dehydration to the microbial inoculums leading to toxicity to cell.
2. The gel retardation assay is a sensitive technique for studying protein–DNA interactions. It is used to study the kinetics of protein DNA interactions. This method relies on the stability of protein DNA complexes when subject to nondenaturing polyacrylamide gel electrophoresis. The DNA is radiolabeled to enable rapid detection and in its native state it migrates quickly through the gel matrix. Protein binding generates slower mobility protein DNA complexes that resolve as discrete bands.
3. Heterologous protein expression depends upon good codon biasing, transcriptional efficiency and optimum copy number production of the expression vector system.
4. Degeneracy produces the ultimate amino acid combination for protein expression.
5. The arrest of thymidine kinase synthesis occurred after completion of the delayed mitosis.
6. For sequencing we need complementary forward and reverse primer to bind with target sequence.
7. Each groove in DNA is lined by potential hydrogen-bond donor and acceptor atoms that enable specific interactions with proteins (seeFigure 27.7). In the minor groove, N-3 of adenine or guanine and O-2 of thymine or cytosine can serve as hydrogen acceptors, and the amino group attached to C-2 of guanine can be a hydrogen donor. In the major groove, N-7 of guanine or adenine is a potential acceptor, as are O-4 of thymine and O-6 of guanine. The amino groups attached to C-6 of adenine and C-4 of cytosine can serve as hydrogen donors.
8. SNP generates single nucleotide polymorphism when RFLP gives the restriction length polymorphism.
9. F-plasmids play a crucial role because they contain partition genes that promote the even distribution of plasmids after bacterial cell division. The bacterial artificial chromosome's usual insert size is 150-350 kbp. A similar cloning vector called a PAC has also been produced from the bacterial P1-plasmid.
10. Allelic exclusion has been observed most often in genes for cell surface receptors and has been extensively studied in immune cells such as B lymphocytes. In B lymphocytes, successful heavy chain gene rearrangement of the genetic material from one chromosome results in the shutting down of rearrangement of genetic material from the second chromosome. If no successful rearrangement occurs, rearrangement of genetic material on the second chromosome takes place. If no successful rearrangement occurs on either chromosome, the cell dies. As a result of allelic exclusion, all the antigen receptors on an individual lymphocyte will have the same amino acid sequence in the variable domain of the heavy chain protein. As the specificity of the antigen receptor is modulated by the variable domain of the light chain encoded by one of the immunoglobulin light chain loci, the specificities of B cells containing the same heavy chain recombination event can differ according to their light chain recombination event.
11. Chlorsulfuron is a selective systemic herbicide, absorbed by the leaves and roots and moves rapidly through the plant. Chlorsulfuron blocks acetolactate synthase and thus prevents the plant from producing a branched chain amino acids. This inhibits cell division in the root tips and shoots of sensitive plants.
12. β-Lactoglobulin is the major whey protein of cow and sheep's milk (~3 g/l), and is also present in many other mammalian species; a notable exception being humans. Its structure, properties and biological role have been reviewed many times.
13. The pancreas is a glandular organ in the digestive system and endocrine system of vertebrates. It is both an endocrine gland producing several important hormones, including insulin, glucagon, somatostatin, and pancreatic polypeptide, and a digestive organ, secreting pancreatic juice containing digestive enzymes that assist the absorption of nutrients and the digestion in the small intestine. These enzymes help to further break down the carbohydrates, proteins, and lipids in the chyme.
14. The recovery after four steps
$$= (0.8)^4 = 0.4096 \simeq 0.41$$
15. Continuous change in the substrate concentration in bulk liquid does not allow enzyme to attain the reverse feedback condition.

16. It creates the substrate based higher influx leading to a positively charged substrate and a negatively charged product.

17. The Ti plasmid contains most of the genes required for tumor formation. Wounded plants exude phenolic compounds that stimulate the expression of the virulence genes(vir -genes), which are also located on the Ti plasmid. The vir genes encode a set of proteins responsible for the excision, transfer and integration of the T-DNA into the plant genome. The genes in the T-DNA region are responsible for the tumorigenic process.

18. Positional cloning is a method of gene identification in which a gene for a specific phenotype is identified only by its approximate chromosomal location (but not the function); this is known as the candidate region. Initially, the candidate region can be defined using techniques such as linkage analysis, and positional cloning is then used to narrow the candidate region until the gene and its mutations are found. Positional cloning typically involves the isolation of partially overlapping DNA segments from genomic libraries to progress along the chromosome toward a specific gene.

19. Auxin helps in generation of the root while cytokinin helps in shoot generation.

20. Reverse genetics is an approach to discover the function of a gene by analyzing the phenotypic effects of specific gene sequences obtained by DNA sequencing. This investigative process proceeds in the opposite direction of so-called forward genetic screens ofclassical genetics. Simply put, while forward genetics seeks to find the genetic basis of a phenotype or trait, reverse genetics seeks to find what phenotypes arise as a result of particular genetic sequences.

21. Fed Batch process is based on feeding of growth limiting nutrient substrate to a culture. This process is used to control repressive effects of the particular nutrient which at higher concentration inhibit the growth of the culture or product.

22. Ecoli doesn't have any splicing system so that it is difficult to achieve the human protein purity in E.Coli cells.

23. Heat sterilization likely the steam sterilization is helpful in large scale production.

24. Embryo Rescue is one of the earliest and successful forms of in-vitro culture techniques that is used to assist in the development of plant embryos that might not survive to become viable plants. Embryo rescue plays an important role in modern plant breeding, allowing the development of many interspecific and intergeneric food and ornamental plant crop hybrids. This technique nurtures the immature or weak embryo, thus allowing it the chance to survive. Plant embryos are multicellular structures that have the potential to develop into a new plant. The most widely used embryo rescue procedure is referred to as embryo culture, and involves excising plant embryos and placing them onto media culture. Embryo rescue is most often used to create interspecific and intergeneric crosses that would normally produce seeds which are aborted. Interspecific incompatibility in plants can occur for many reasons, but most often embryo abortion occurs In plant breeding, wide hybridization crosses can result in small shrunken seeds which indicate that fertilization has occurred, however the seed fails to develop. Many times, remote hybridizations will fail to undergo normal sexual reproduction, thus embryo rescue can assist in circumventing this problem.

25. Organic ingredients intended for composting can alternatively be used to generate biogas through anaerobic digestion. Anaerobic digestion is fast overtaking composting in some parts of the world including central Europe as a primary means of downcycling waste organic matter.

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